

**Team:** Slugriculture

**Institution:** University of California, Santa Cruz

**Division:** Division I (4-year universities and colleges)

### Understanding this Report:

Across all Farm Robotics Challenge (FRC) divisions, within each judging round, entries received a scoring review from several judges. FRC judging panels consist of multidisciplinary teams with academic, industry, and finance backgrounds.

This year, FRC invited judges to provide feedback on each scoring criteria. To aid your

interpretation of judges' feedback, we have applied a simple round and number designation for each judge (e.g., Preliminary Round Judge 1).

Additionally, each scoring criterion has had its 'weight' noted (e.g., Problem Identification [10% weighted score]). While each criterion is critical to a project's success, FRC places a heavy emphasis on design and overall project impact.

### Problem Identification [10% weighted score]:

*Evaluates how clearly the team defines a real agricultural problem and demonstrates understanding of its context and constraints.*

Preliminary Round Judge 1:

Cost-effective solution for target customer (small farmers) to identify crop damages and pest outbreak, with crops such as artichokes.

Preliminary Round Judge 2:

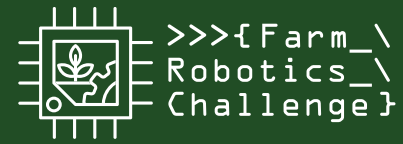
Needs for ground scouting is well stated and defined

Finalist Round Judge 1:

The project identifies a highly relevant and often overlooked bottleneck in agricultural AI deployment: scalable dataset annotation and field-label generation. The report clearly explains how manual labeling, delayed scouting, and limited technical resources constrain adoption of management tools, particularly for smaller growers. The framing is thoughtful and operationally grounded. The deduction is primarily because the report focuses heavily on annotation efficiency but does not fully address the broader challenge of timely intervention. In many agricultural disease and pest systems, visual signs of infection are often detected too late for effective mitigation, which remains a major limitation even with commercial multispectral imaging platforms such as MicaSense and Sentera systems. While the proposed workflow improves dataset generation and model development, it is less clear how it fundamentally improves early detection timing under real operational conditions.

Finalist Round Judge 2:

BirdsEye targets a specific and well-understood pain point: specialty crop growers in coastal California discover pest and disease pressure late, after manual walkthroughs have already missed the intervention window. What elevates this above a typical academic framing is the core technical insight — that the bottleneck to AI adoption in ag isn't the AI itself, it's the labeled training data. That reframe is sharp and comes from actual field experience. Minor deduction: the primary validated case study (burrowing mammal detection) is a somewhat unusual entry point for



a crop monitoring platform, and the strawberry/artichoke pilot studies — which are more commercially compelling — are still early-stage.

Finalist Round Judge 3:

Strong "customer-centric" problem statement framed from a small-farmer perspective: discovering pest/damage problems too late because of labor-intensive walkthroughs. The team goes deeper than "monitoring is hard" by identifying the labeled-training-data bottleneck as the underlying reason AI-driven monitoring hasn't reached small farms — a real and well-argued root cause.

Finalist Round Judge 4:

I get that the idea is that AI and precision agriculture is difficult, but I'm not clear on what this actually does / how it integrates with other solutions to enable the precision ag. Also not clear what crops this applies to, what diseases, what it's for. If covers everything for everything, they should be more clear and provide a few concrete examples.

Finalist Round Judge 5:

The team clearly identifies a massive, real-world bottleneck in smart agriculture: manual frame-by-frame annotation of aerial drone imagery. They frame the problem beautifully around small-scale farmers who cannot afford dedicated technical machine learning teams but face severe crop losses. But the problem statement heavily relies on a qualitative user story quote and I think the project lacks on early high-level financial figures or regional crop loss statistics to quantify the regional economic scale of pest outbreaks prior to the design explanation.

Finalist Round Judge 6:

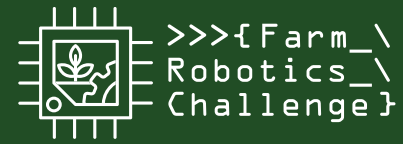
The report clearly defines the agricultural problem of delayed pest and disease detection while demonstrating a good understanding of labor constraints, affordability barriers, field-operational realities, and the technical limitations of existing precision-agriculture systems.

Finalist Round Judge 7:

(+) The team identifies a clear and relevant agricultural problem: small and mid-scale growers often detect pests, disease, and crop damage too late because manual scouting is labor-intensive, incomplete, and inconsistent. (+) The project correctly recognizes that the lack of labeled, farm-specific training data is a major barrier to practical AI adoption in agriculture. This is a thoughtful and important framing because it addresses not only detection, but the data bottleneck that prevents many AI systems from becoming useful on farms. (+) The problem is framed around affordability and accessibility, which is especially relevant for smaller farms that may not be able to use expensive commercial precision-agriculture systems. (+) The report connects the problem to multiple real use cases, including burrowing mammal damage, two-spotted spider mite stress, crown rot, and general crop-health monitoring. (-) The report would benefit from stronger quantitative framing of the problem's economic magnitude at the outset, such as estimated annual losses from delayed detection in the target crop segments. (-) The initial problem statement focuses most clearly on burrowing pests and general crop stress, while TSSM and crown rot are introduced later. The framing would be stronger if these applications were more explicitly connected to the original problem definition.

Finalist Round Judge 8:

The problem statement is generic and framed in terms of technical challenges. The argument is that you want to help small farmers with tagged data image sets at small farmer economics. I am going to push back on that



segment, and I think if you talk to the growers behind the PCA you would likely determine that small growers are under the most margin pressure and have the least available revenue to spend on AgTech solutions of any sort. I also think if you talked to most growers, they would say they are not tech wizards in many cases and many (i.e. most of them that are not early adopters) would likely prefer an as-a-service or rental model type of business purchase option over buying low cost sensors and drones and doing some assembly work and fairly in-depth technical integration. Anything that's not relatively plug and play will be tough to get widescale adoption of for the small farm segment.

### **Ag Partner Collaboration [10% weighted score]:**

*Evaluates the extent to which the team engaged with an agricultural partner to inform problem selection, design decisions, and validation.*

Preliminary Round Judge 1:

Strong evidence of collaboration with the pest control advisor, as well as with Rawlings, Lawson et al.

Preliminary Round Judge 2:

The team closely worked with the Ag partner to use the partners field to collect data.

Preliminary Round Judge 3:

The team closely worked with the Ag partner to use the partners field to collect data.

Finalist Round Judge 1:

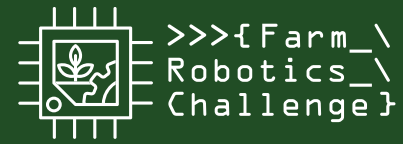
Strong collaboration with PCAs, growers, and agricultural stakeholders is evident throughout the report. The team clearly incorporated real-world workflow considerations into the system design and deployment strategy. However, some operational aspects remain underdefined, particularly how RTK workflows integrate practically into normal scouting operations for field advisors and operators so that field observations can be used for label creation.

Finalist Round Judge 2:

Five partners, not one, and each shaped a different dimension of the system. Joe Aguilar (Zap Drone Services) drove the pest detection focus and real-field deployment approach. Greg Rawlings (Jacobs Farm del Cabo) shaped the GPS pipeline. Jered Lawson (Pie Ranch) held the team accountable on cost and accessibility — and you can see that pressure reflected in the \$5,751 total system cost. Dr. Waterhouse provided a structured testbed at UCSC Farm for benchmarking against commercial platforms. This is the kind of multi-partner engagement where feedback visibly changed design decisions rather than just appearing in the acknowledgments.

Finalist Round Judge 3:

Five distinct, complementary ag partners: Joe Aguilar (Pest Control Advisor, Zap Drone Services — pest pressure field access), Greg Rawlings (Jacobs Farm del Cabo — burrowing-mammal trials, GPS workflow input), Jered Lawson (Pie Ranch — end-user adoption and cost emphasis), Dr. Hannah Waterhouse (UCSC Environmental Studies — cover crop testbed), and Dr. Jon Detka (CSUMB WATRS — multispectral expertise). Each partner shaped a different aspect of the design — the breadth and specificity are exemplary.



Finalist Round Judge 4:

They have a farmer in the video, but not clear why this solution solves what their concern is or what feedback was given in the design, etc.

Finalist Round Judge 5:

Exceptional depth and breadth of partnerships

Finalist Round Judge 6:

The project demonstrates a strong engagement with growers, pest control advisors, researchers, and agricultural partners whose feedback directly shaped the system's deployment, affordability goals, workflows, and validation studies.

Finalist Round Judge 7:

(+) The team worked with multiple relevant partners, including both end users and technical/domain experts. This gives the project a strong practical foundation.(+) Partner contributions are documented with useful specificity, particularly around field access, scouting needs, affordability, and the practical constraints of farm monitoring.(+) The combination of grower input and technical/domain expertise strengthens the project because the team is not designing only for a technical benchmark; it is designing around real field workflows.(+) The partners helped ground the project in several agricultural settings rather than a purely laboratory or demonstration environment.(-) The depth of ongoing partner engagement during the evaluation phase is less clear. It is not fully evident whether partners reviewed interim outputs, validated the detections, or directly assessed the final workflow. (-) The report notes that season-long economic data could not be obtained from farm partners. This is understandable, but it limits the ability to validate the impact and ROI claims.

Finalist Round Judge 8:

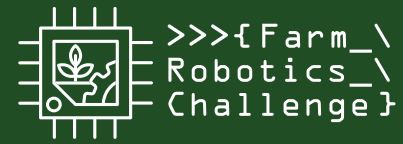
The PCA conversation is a good first step, but I think you need to get to farmer conversations (which most of the other teams I have reviewed so far have managed to do) and it needs to be small farmer conversations. Then you need to find out how they get tagged image data now and what it costs them and the value they get from it. Then the challenge is delivering more value (without too much incremental risk from testing and integrating a new solution) at a price that is better (it can't just be a tie or there's not enough savings to move down the path of an entirely new technology often from an unproven startup that could cease to exist in the next 12-18 months. Virtually every farmer has had a situation where the technology or the startup was purchased or shut down and the product is no longer supported. Those scars last a long time in farmer's memories. To really round out the collaborative element, a small grower needs to be in the conversation (preferably talking about some math).

### **Overall Technical Design [20% weighted score]:**

*Evaluates the quality, feasibility, and integration of the system's technical components.*

Preliminary Round Judge 1:

Clear operational procedure is noted (having a very clear 5-step process is excellent). Diagram (Fig 1) shows the different aspects of the system. Technical details shown in Figure 2 with adequate detail. Hardware platform and software backend sections have necessary depth but are portrayed in a concise and understandable manner. Six



performance factors noted, one elaborated on with a plot. Extremely strong on safety (5 parts, all useful and important). Good list of cost of materials (Tables 1-3).

### Preliminary Round Judge 2:

The team argues that the drone can fly up to 30 minutes without a payload. (Page 7 Custom UAV Build). Making a 30-minute flight drone is very challenging. Is it actually tested? Conventional custom drones usually fly for around 15 minutes with ordinary setups. The report says the drone flight duration was 7 min 1 sec. It would be great if they had more detailed flight test results, such as average current and the amount of battery mAh consumed.

### Finalist Round Judge 1:

The system architecture is technically mature and well-integrated. The use of RTK georeferencing, multispectral sensing, ROS2 infrastructure, synchronized image acquisition, automated annotation pipelines, and geospatial reconstruction demonstrates strong engineering capability. However, the project is more compelling as an AI/data infrastructure platform than as a robotics innovation. The UAV implementation itself relies largely on established aerial surveying workflows, while the strongest innovation lies in the geospatial annotation framework, workflow orchestration, and AI pipeline automation.

### Finalist Round Judge 2:

The system architecture is coherent and well-integrated: RTK geoannotation, multispectral + RGB payload, offline batch processing compatible with consumer laptops, and a software pipeline that runs on open-source tools. The insight of shifting annotation from screen to field — and pairing it with 3D reconstruction — is genuinely novel. The custom UAV build at \$4,493 with DJI M300 performance comparisons is well-documented. Gaps: the system has not been tested end-to-end at commercial scale, and the hardware constraint (Raspberry Pi-based payload) introduces processing limitations the team acknowledges but has not yet quantified in real-time field conditions.

### Finalist Round Judge 3:

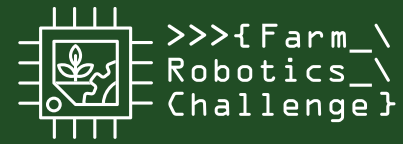
Genuinely novel approach to the labeled-data bottleneck. Custom UAV + payload (Raspberry Pi 5 + 8 MIPI cameras across RGB and band-filtered monochromatic multispectral + AS7265x spectrometer + radar altimeter + RTK GPS) plus base station and handheld geoannotation rover. Software pipeline (OpenDroneMap, NerfStudio, YOLO26-seg) is sophisticated. The RTK NTRIP server design that supports Amiga is a thoughtful cross-platform concession. However, the core platform is an aerial system that requires Part 107 piloting for legal commercial use, plus multi-stage workflow training (base setup, ground annotation, flight planning, post-processing). For the team's stated target — small farms — that operator-accessibility gap is significant. Engineering originality is high; the appropriate-technology-for-the-task criterion is moderated.

### Finalist Round Judge 4:

Design seems good enough.

### Finalist Round Judge 5:

The hardware/software pipeline is exceptionally well thought out and innovative. Automatically transforming GPS-tagged ground observations into image-space coordinates via projection models completely eliminates manual labeling. The custom UAV build offers high payload capabilities (3kg) with open-source infrastructure. Cons are: the hardware architecture relies on a Raspberry Pi 5 to drive 8 MIPI cameras, handle serial communications, and manage multi-sensor data collation. Processing 8 visual streams simultaneously on a Pi 5 may create immediate computational and thermal throttling vulnerabilities in harsh field conditions.



### Finalist Round Judge 6:

The report presents a sophisticated and integrated precision-agriculture monitoring platform combining UAV robotics, RTK annotation, multispectral imaging, AI dataset generation, 3D reconstruction, and autonomous detection pipelines for a low-cost system. In short, the system physically looks like a set of Nintendo parts and Home Depot buckets glued together, but in reality it represents a nice step forward in affordable drone remote sensing technologies.

### Finalist Round Judge 7:

(+) The system architecture is well integrated across hardware and software layers, combining UAV sensing, RTK localization, field annotation, automated labeling, AI model training, and geospatial outputs. (+) The geoannotation system is the project's most distinctive technical contribution. By linking field observations to aerial imagery, the team directly addresses the manual labeling bottleneck that often prevents agricultural AI from being deployed practically. (+) The technical choices are consistently justified with respect to cost and accessibility. The use of off-the-shelf components, open-source software, and low-cost compute reflects a realistic design philosophy for smaller farms and service providers. (+) The system has a clear path toward multiple outputs from one flight: AI training data, detections, 3D reconstructions, multispectral maps, and geospatial field intelligence. (+) For the Drone Applications category, the design is particularly strong because the UAV is not merely collecting images; it is embedded in a broader workflow for farm-specific data generation and decision support. (-) The workflow still appears technically complex for non-expert farmers, especially RTK setup, field tagging, UAV mission planning, postprocessing, and AI model training. (-) The image resolution of 1280x800 at 1–10 Hz may limit performance for fine-scale targets such as early-stage TSSM infestation. A clearer discussion of resolution requirements for each detection task would strengthen the design. (-) The reliance on offline batch postprocessing is practical for the current prototype, but it introduces latency between data collection and actionable output. The pathway toward near-real-time processing for time-sensitive pest management remains unclear.

### Finalist Round Judge 8:

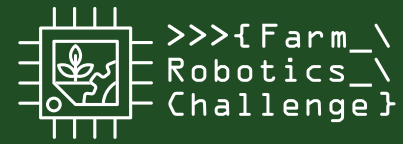
I think the technical design is solid but I think it focuses on the tech too much and not enough on the user experience capabilities of the average small farmer user. Most of them are not going to know what open source is, are not going to be interested in crowd sourcing data, and will likely get confused if they even watch the video. It needs to be simpler for the average small farmers - the early adopters are only 3-5% of the small farmer target audience and everyone else in the segment will need to be offered a much simpler to implement and integrate solution. I'm not sure how you can pivot to that model from where you are at now. You would need a service layer over the top of the current product to hide the complexity behind a human interface that delivers the solution instead of requiring the farmer to build it or partially integrate it.

### **Design Evaluation [15% weighted score]:**

*Evaluates how effectively the team measures and demonstrates system performance using relevant metrics and data.*

### Preliminary Round Judge 1:

Three metrics noted (early detection accuracy, operational efficiency, and cost savings) explained with great depth. Relevance is noted for each one, details on the studies' purpose and procedure, the type of data that was collected, and the results, along with a summary of the key findings and implications. All three metrics are important, but



very good pick on #2 (time savings), and the result of 35x (max) is very promising. Cost is 30-90% better than alternatives, which is also a strong point, and important for target clients.

### Preliminary Round Judge 2:

The team completed multiple end-to-end tests and proposed several metrics. But it would be great if they had; more evaluations on drone flight evaluations since they suggested the drone is one of the main product location error metrics, between ground scouted vs aerial detected (Figure 1 shows the visualization of the errors, but not in metrics).

### Preliminary Round Judge 3:

Solid evaluation section overall, the independent test dataset and cross-site validation are genuinely strong choices. That said, the AI detection component felt a bit vague compared to the rest. Claims like 'successfully generalized' and 'demonstrated robustness' would land better with harder numbers behind them. With 2,000+ test images you have the data to be specific, how many failed, and under what conditions? Precision would also be worth including alongside recall, since without it it's hard to know how often the system is crying wolf, which matters a lot for whether a farmer would actually trust and use it.

### Finalist Round Judge 1:

The project includes meaningful field validation, operational timing comparisons, and cross-location testing. The annotation automation workflow and recall metrics are credible when presented. However, some evaluation sections lack sufficient rigor or clarity. Performance metrics for the TSSM detection workflow are not clearly presented, limiting evaluation of practical deployment readiness. Additionally, reported time savings appear somewhat overstated because UAV survey time does not appear to include downstream image processing, analysis, and report-generation. Which are necessary steps before actionable pest recommendations can be delivered.

### Finalist Round Judge 2:

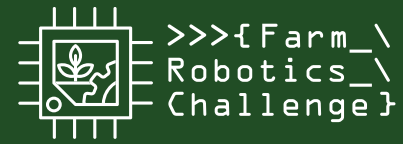
84% recall on burrowing mammal detection on an independent test dataset 20 miles from training sites is a strong result, and the 44,000+ labeled images generated with minimal manual effort validates the core annotation pipeline. The 22-35x efficiency gain over manual scouting is well-supported by paired timing data. The TSSM and crown rot pilot studies demonstrate early-stage sensitivity to the use cases that matter most commercially, though these remain feasibility demonstrations rather than validated detection benchmarks. The team was honest about scope reductions from the original plan, which adds credibility.

### Finalist Round Judge 3:

Three rigorous evaluation metrics, each well-grounded: 84% recall on burrowing-mammal detection on an independent test set 20 miles from training (real generalization, not in-distribution overfit), 7x–35x operational efficiency vs. manual scouting at 25 ft/min in cover-crop monitoring (with explicit RTK-setup vs flight-only breakdowns), and a quantitative cost comparison (\$5,751.76 vs. \$8,128 DJI Mavic 3M / \$33,203 DJI M350 / \$43,840 IF-800). Two pilot studies (TSSM, crown rot) demonstrate generalizability beyond the burrowing-mammal case study.

### Finalist Round Judge 4:

No metrics given here.



### Finalist Round Judge 5:

Excellent quantitative benchmarking against leading market alternatives (DJI Mavic 3M, M350 RTK, etc.) But the pilot study datasets for critical threats like Spider Mites and Crown Rot are quite thin, relying on only 4 flights each. The report also lists high-level qualitative findings like "canopy discoloration" (p. 17) but fails to provide standard AI accuracy metrics (precision, recall, mAP) for these two diseases.

### Finalist Round Judge 6:

The project thoroughly demonstrates system performance using recall metrics, timing comparisons, scalability analysis, multispectral validation studies, dataset statistics, and cost benchmarks against commercial alternatives.

### Finalist Round Judge 7:

(+) The primary case study on burrowing mammal detection is methodologically strong and gives the project a credible field validation anchor. (+) The operational efficiency evaluation is well structured. The paired timing comparison between manual and UAV scouting, including RTK setup overhead, gives a practical view of how the system could change field-monitoring workflows. (+) The team generated more than 44,000 labeled training images, which is a strong demonstration that the geoannotation workflow can scale dataset creation. (+) The cost comparison is compelling, with a full system cost of \$5,751.76, lower than several commercial UAV alternatives. (+) The independent test-site result is important because it provides some evidence of generalization beyond the exact field where the model was trained. (-) The strongest quantitative detection results are for burrowing mammal holes, not directly for crop pest or disease detection. (-) The TSSM and crown rot studies remain at the pilot or feasibility stage. Quantitative performance metrics such as precision, recall, and F1 score are not reported for these use cases, which limits confidence in readiness for crop pest/disease deployment. (-) Precision is not reported alongside recall in the primary case study, making it difficult to assess the false-positive burden that growers would face in practice. (-) The claim of "earlier detection" is plausible, but it has not yet been demonstrated across a crop production cycle.

### Finalist Round Judge 8:

As above, the presentation and narrative got too focused on technology elements and not focused enough on actual grower problems. I think the metrics would need to be around grower adoption and potentially of a different as-a-service model. I just have not seen small farmers adopting open source or crowd sourcing data sets with any frequency at all and many have tried. Not sure if business model falls under design but that's where the disconnect is at the moment.

## **Project Story [10% weighted score]:**

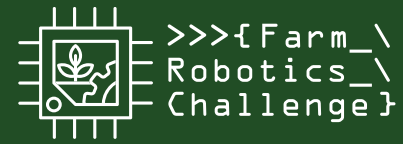
*Evaluates how clearly the team communicates their design process, decisions, challenges, and learning.*

### Preliminary Round Judge 1:

An encountered problem was solved by this system with feedback from partners. Challenges were noted and solved, albeit by compromise.

### Preliminary Round Judge 2:

Story is great; 1. Ground scouting is needed for targeted image collection with a UAV. 2. So the team made a system with a UAV, RTK base, and a clicker.



### Finalist Round Judge 1:

The report is very coherent and professionally structured. The team clearly communicates the evolution of the project and demonstrates strong systems-level thinking. The report also does a good job acknowledging practical annotation ambiguity and operational limitations. The strongest aspect of the narrative is the consistent focus on solving one central operational bottleneck rather than adding unnecessary complexity.

### Finalist Round Judge 2:

Clear origin story grounded in real frustration with the labeling bottleneck, articulated before pitching the solution. The narrative connecting UCSC HARE Lab experience → grower partner feedback → design decisions is logical and traceable. The team's decision to scope down from 40+ field trials to two high-impact case studies rather than produce thin results across a wider set shows good judgment. The challenge-and-adaptation section is one of the more honest in this cohort.

### Finalist Round Judge 3:

Honest narrative about the manual-labeling bottleneck observed in the UCSC HARE lab leading to the BirdsEye approach. Scope was adjusted from 40 to 26 flights and from full economic season-long evaluation to system-cost demonstration, with rationale explicitly explained. Shows iteration and adaptation under realistic constraints; the burrow-definition variance discussion (Figure 6) is a thoughtful design-honesty moment.

### Finalist Round Judge 4:

No discussion of tradeoffs, decisions, challenges.

### Finalist Round Judge 5:

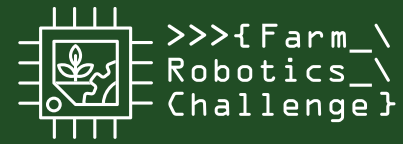
The team provides a strong, organic origin story. The report outlines how specific expert feedback directly shaped the physical system architecture. Bad comments: The story skips straight from identifying the problem to the completed dual-RTK drone solution. It fails to document the iterative design failures or early prototypes of the handheld geoannotation clicker unit before settling on the final Wio Terminal hardware

### Finalist Round Judge 6:

The project report communicates the project's design process very clearly through detailed system explanations, engineering rationale, workflow diagrams, validation strategy, and candid discussion of scope limitations and design tradeoffs.

### Finalist Round Judge 7:

(+) The team tells a clear and thoughtful story about identifying manual image labeling as a major bottleneck in agricultural AI and designing BirdsEye to reduce that barrier.(+) The project narrative connects technical development to real partner needs, especially cost, accessibility, and practical field workflow.(+) The team is transparent about adapting its scope, which gives the report a credible and reflective tone.(+) The story is especially effective when it explains why a low-cost UAV and geoannotation workflow could make advanced monitoring more accessible to smaller farms.(-) The discussion of technical challenges encountered during development is relatively brief. More detail on specific engineering obstacles and how they were resolved would strengthen the narrative. (-) The transition between design iterations is not fully articulated. The report would benefit from a clearer account of what the team learned from early prototypes and how those lessons shaped the final system.



Finalist Round Judge 8:

Recurring theme - too many tech topics, not enough grower problem topics. Get really focused on the potential problem the new image solution with some auto-tagging/user tagging capabilities can solve. What does the current solution cost and what is it lacking? Then how do you tie your solution into that gap with economics that are competitive?

### **Overall Project Impacts [35% weighted score]:**

*Evaluates the project's real-world relevance, effectiveness, and potential for practical or commercial application.*

Preliminary Round Judge 1:

Shows that performance exceeded expectations on all 3 metrics, which provide the usefulness of this system to solve the problem; there is strong commercial potential; environmental and social impacts noted.

Preliminary Round Judge 2:

I think the clicker has the most impact on real-world applications. However, it could be done without re-inventing drones. TLDR: Making a home-made drone is good for saving money on research, but it needs more consideration when comparing prices with commercial drones. For example, I would sell the current \$6K BOM drone for \$12K or more, including all labor and after-service. Long version: They compared the drone's price point, which is much cheaper than that of a commercial drone. But even if the cost of a commercial drone with similar specs is way more expensive, the actual material cost may be similar or even less because of mass production; the final cost of the product not only contains BOM but also contains a lot of other things, such as R&D, marketing, salary, investment, etc. The base station faces this dilemma as well.

Preliminary Round Judge 3:

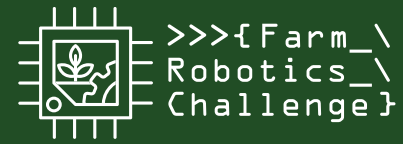
I think the problem you guys addressed is incredibly important, excellent job. I hope to see more research focused on helping small farmers like you do here.

Finalist Round Judge 1:

The project has strong potential as a scalable agricultural AI infrastructure tool, particularly for research groups, PCAs, cooperatives, and service providers. The low-cost and modular philosophy is commendable. However, the commercialization pathway for direct small-grower adoption is less convincing. In practice, systems requiring UAV operation, RTK workflows, and multispectral imaging infrastructure may be adopted more commonly through service providers, extension programs, or cooperatives rather than individual owners by small farms.

Finalist Round Judge 2:

The impact case is real: earlier detection, reduced scouting labor, AI access for small farms that previously couldn't afford it. The cost comparison (\$5,751 vs. \$33,000-\$43,000 for commercial equivalents) is credibly documented. The commercial pathway is less developed — the team identifies the right customer segments but doesn't quantify the market or articulate a business model beyond the platform itself. The environmental benefit (targeted treatment → reduced chemical use) is mentioned but not quantified.



### Finalist Round Judge 3:

Same fundamental commercial-scaling challenge as HydroFleet — an aerial platform whose adoption pathway requires Part 107 pilots, base-station infrastructure, and offline post-processing literacy that small farms (the team's stated target) do not have. The "stop-when-you're-happy" modular tiers help on paper, but the highest-value tier — the AI-detection-flywheel tier — still requires aerial operation. The cost comparison (\$5,751 vs. \$8,128 DJI Mavic 3M / \$33,203 DJI M350 / \$43,840 IF-800) benchmarks against other UAV systems rather than against grower-deployable ground equipment, which understates the real adoption barrier. The "turns data labeling into a non-issue" insight is genuinely powerful, but the value mostly accrues to agronomy service businesses and researchers rather than to growers directly. As a commercial product targeted at the team's stated small-farmer beachhead, the platform faces structural headwinds that the modular tiers cannot fully resolve.

### Finalist Round Judge 4:

I think I get what they're trying to do as far as this being an AI play that enables precision agriculture, but not sure I get whether it's doing it all on its own or if it needs to integrate with other products, what the targets are, how to evaluate savings, etc.

### Finalist Round Judge 5:

High academic honesty. The team transparently addresses their failures.

### Finalist Round Judge 6:

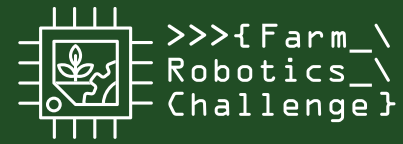
As a remote sensing specialist myself, I can't help appreciate how this project demonstrates exceptional real-world relevance by addressing major barriers to AI adoption in agriculture through an affordable, scalable, and field-validated precision-monitoring platform for small and mid-scale farms. This technology could have saved me countless hours over my years of field research using drones. However, for the average farmer, this technology may still be a little outside their direct necessities. For that reason alone, I marking this section as a 7, while as a remote sensing expert, I want to rank it as a 10.

### Finalist Round Judge 7:

(+) The project has strong potential impact because it addresses a major barrier to precision-agriculture adoption: the cost and expertise required to collect labeled data and deploy AI tools.(+) The low-cost system design could make advanced UAV monitoring more accessible to small and mid-scale farms.(+) The "beyond-human consistency" result, where the model identified targets missed by human annotators, is a notable finding and strengthens the case for AI-augmented scouting.(+) The system could support reduced chemical use by enabling earlier and more targeted treatment, especially if the pest and disease detection modules are further validated.(-) Impact claims related to chemical-use reduction, crop-loss prevention, and ROI within a single growing season are not yet supported by field data. These claims are plausible but would benefit from preliminary empirical grounding.(-) The commercial pathway and adoption strategy are outlined at a high level but need more specificity regarding deployment readiness, regulatory constraints for UAV operations at scale, and the learning curve for non-technical operators.(-) The strongest demonstrated use case is still burrowing mammal detection; broader crop-health impacts remain promising but less fully validated.

### Finalist Round Judge 8:

I think the biggest challenge for this project is the mismatch between target market segment (small farmers - often low-tech small farmers) and an imaging solution that requires the users to execute a training effort, then collect the data, then tag it, then crowd source and share the data so you can load other data, and then try and make sense of



the crowd sourced data as a complete data set with third party data added in. Again, if you could mask the complexity of the crowd sourcing and just show it as a benefit to the user, that would create a much bigger market opportunity.

### **Judges' Final Comments Regarding Entry:**

*Some judges elected to comment on the overall project, separate from any scoring criterion.*

#### Preliminary Round Judge 1:

The strongest point of this system is the development of the system (the bucket for greater GPS accuracy, the handheld device, and the drone) in-house in a modular approach that decreases the cost significantly and makes it viable for small farmers. Clear time savings of somewhere between 10x and 22x, which is extremely significant.

#### Finalist Round Judge 3:

14-person team (Slugiculture) at UC Santa Cruz, advised by Steve McGuire (Assistant Professor of Electrical and Computer Engineering). The breadth (electrical/mechanical/computer science) supports the multidisciplinary hardware-plus-AI scope. The team's most distinctive choice is to attack a root-cause problem (labeled-data bottleneck) rather than a symptom, and the submission demonstrates that approach can produce a low-cost, generalizable, and broadly applicable platform.

#### Finalist Round Judge 6:

This project is innovative, technically rigorous, and solidly grounded in real challenges of agricultural monitoring with drones. It does an outstanding job combining advanced robotics, AI, and practical field deployment into a scalable and accessible precision-agriculture platform. The only two downsides of the project are that the technology does not look polished (for users to get excited about adopting it), and that it isn't quite as directly applicable to defined agricultural problems as some of the other projects within this competition. That said, as a remote sensing specialist, I have a great appreciation for this project!

#### Finalist Round Judge 7:

BirdsEye is a strong and well-motivated project that tackles a genuine bottleneck in agricultural AI: the difficulty of generating labeled, farm-specific training data. The system's integration of UAV sensing, RTK geoannotation, automated labeling, and field-scale mapping is impressive, and the cost/accessibility framing is well aligned with the needs of small and mid-scale farms. The main limitation is that the strongest quantitative validation is for burrowing mammal detection, while the crop pest and disease studies remain more exploratory. Precision is also not reported for the primary detection case, making it harder to assess false-positive burden for growers. Even with these limitations, this is the cleanest fit for the Drone Applications prize if/once HydroFleet is selected for the Grand Prize (again I didn't have the chance to review the UCD submission due to COI). BirdsEye uses the drone as the core enabling platform for agricultural intelligence, not simply as a camera carrier.